多莉：一只羊的复制

Dolly, the Copied Sheep



白脸小羊为何由黑脸母羊生下?

英文原文

Dolly was born at Roslin, **Scotland**, on 5 July 1996. Her **genome** came from a Finn Dorset ewe's mammary cell, **fused with** a Blackface egg **stripped of** its **nucleus**. A Blackface **surrogate** carried the **embryo** through **pregnancy**, yet her white face was **proof of** a **genetically** different **lineage**. She alone survived from 277 attempts, an **extraordinary** failure rate.

Dolly is **popularly** called the first **clone**, a **misconception**. Her **predecessors** included frogs cloned from **transplanted** nuclei in 1962 and Roslin's embryo-derived **duo**, Megan and Morag, in 1995. She was **genuinely** the first **mammal** cloned from an adult cell. Roslin's **announcement** on 22 February 1997 **coincided with** the paper, and her name, a mammary **pun**, **honoured** Dolly Parton.

The news **provoked uproar** and **apprehension** about the **ethics** of human cloning. In March 1997 Clinton **prohibited** federal funding for such experiments. In 1999 her telomeres, or **chromosome** tips, were reported **shortened** by about 20% for her age. That fed **anxiety**; some **speculated** she was ageing **prematurely**.

Dolly **was put down**, aged six, on 14 February 2003, with **arthritis** and a **progressive lung** disease caused by a sheep **virus**. Indoor **confinement** probably **contributed**. In July 2016, 13 clones, four from her cell line, were ageing normally, without **hypertension** or **diabetes**. In November 2017, her **skeletal** X-rays showed ordinary arthritis, and fears of early **onset** were judged **unfounded**.

She bore six lambs by natural **mating**; Bonnie, first of these **descendants**, arrived in April 1998. Since 2003 her **stuffed** body, now **famously** an **icon** and **memorial**, has been **encased in** the National Museum of Scotland. Behind glass, her white face **plainly testifies to** her Finn Dorset **ancestry**.

中文译文

1996年7月5日，多莉在苏格兰罗斯林出生。她的基因组来自一只芬兰多塞特母羊的乳腺细胞，这个细胞融合了一枚黑脸羊卵细胞，这枚卵已去除了细胞核。一只黑脸羊代孕母羊怀着这个胚胎度过了孕期，而多莉的白脸则是一个基因上不同血统的证明。只有她从277次尝试中存活了下来，一个异乎寻常的失败率。

多莉一般被称为第一个克隆体，这是一个错误的观念。她的前几代包括于1962年由移植的细胞核克隆出的青蛙，以及罗斯林于1995年由胚胎衍生的一对绵羊：梅根和莫拉格。她确实是第一个用成年细胞克隆出的哺乳动物。1997年2月22日罗斯林的公告与这篇论文同时发生，而她的名字，一个关于乳腺的双关语，致敬了多莉·帕顿。

这一消息引发了轩然大波，以及关于人类克隆伦理的忧虑。1997年3月，克林顿禁止联邦资金用于这类实验。1999年，她的端粒，即染色体末端，据报道比同龄羊缩短了约20%。这加剧了人们的焦虑；有人推测她正在过早衰老。

多莉被实施安乐死，时年六岁，那是2003年2月14日；她患有关节炎和一种进行性的肺部疾病，引起它的是绵羊病毒。被限制在室内很可能是原因之一。2016年7月，13只克隆羊，其中4只来自与她相同的细胞系，正在正常衰老，没有高血压，也没有糖尿病。2017年11月，她的骨骼X光片显示出寻常的关节炎，而对提前发病的担忧被判定为毫无根据。

她产下六只小羊，是通过自然交配；邦妮是这些后代中的第一只，生于1998年4月。自2003年起，她的填充标本，如今已成为出了名的标志和纪念物，一直封存在苏格兰国家博物馆。隔着玻璃，她的白脸清楚地证明着她的芬兰多塞特祖源。

词汇与表达

Scotland n.
苏格兰

genome n.
基因组
Her genome came

fused with v.
融合了
fused with a Blackface egg

stripped of v.
去除了
stripped of its nucleus

nucleus n.
细胞核
stripped of its nucleus

surrogate n.
代孕母羊
Blackface surrogate

embryo n.
胚胎
carried the embryo through pregnancy

pregnancy n.
孕期

proof of n.
证明

genetically adv.
基因上

lineage n.
血统
a genetically different lineage

extraordinary adj.
异乎寻常的

popularly adv.
一般

clone n.
克隆体
the first clone

misconception n.
错误的观念
a misconception

predecessors n.
前几代

transplanted v.
移植

duo n.
一对

genuinely adv.
确实

mammal n.
哺乳动物
the first mammal cloned

announcement n.
公告

coincided with v.
与……同时发生
coincided with the paper

pun n.
双关语

honoured v.
致敬了

provoked v.
引发了
The news provoked uproar

uproar n.
轩然大波

apprehension n.
忧虑
apprehension about the ethics

词汇与表达（续）

ethics n.
伦理
the ethics of human cloning

prohibited v.
禁止
Clinton prohibited federal funding

chromosome n.
染色体

shortened v.
缩短了

anxiety n.
焦虑

speculated v.
推测
some speculated

prematurely adv.
过早
she was ageing prematurely

was put down v.
被实施安乐死

arthritis n.
关节炎

progressive adj.
进行性的
a progressive lung disease

lung n.
肺部

virus n.
病毒

confinement n.
被限制
Indoor confinement

contributed v.
原因之一

hypertension n.
高血压

diabetes n.
糖尿病

skeletal adj.
骨骼

onset n.
发病
fears of early onset

unfounded adj.
毫无根据
were judged unfounded

mating n.
交配

descendants n.
后代

stuffed adj.
填充标本

famously adv.
出了名

icon n.
标志

memorial n.
纪念物

encased in v.
封存在

plainly adv.
清楚地

testifies to v.
证明

ancestry n.
祖源
her Finn Dorset ancestry

文章简述

- 诞生：研究者将去掉细胞核的卵细胞与成年母羊乳腺细胞融合，培育胚胎，1996年由代孕母羊产下多莉。
- 突破：277次尝试仅多莉存活；她是首只由成年细胞克隆的哺乳动物，此前已能用胚胎细胞克隆羊。
- 争议：成果引发人类克隆伦理争论；多莉的端粒（染色体末端）比同龄羊短约20%，又引起早衰猜测。
- 离世：多莉自然交配育下六只小羊；患有关节炎和病毒性肺病，六岁时被实施安乐死，不能据此认定克隆导致早衰。
- 复核：后续研究中13只克隆羊正常衰老；多莉骨骼复查也未支持关节炎过早出现的担忧。